

1 Problem: Gridworld Navigation

Consider a simple 3×3 Gridworld where an agent navigates to reach a goal state. The states are labeled s_1 through s_9 as shown below:

s_1	s_2	s_3
s_4	s_5	s_6
s_7	s_8	s_9

The agent can take four possible actions: $\mathcal{A} = \{\text{Up, Down, Left, Right}\}$. State s_9 is a terminal goal state with reward $+10$. All other transitions yield a reward of -1 . If the agent attempts to move outside the grid, it remains in the current state and receives the transition reward.

Parameters

- Discount factor: $\gamma = 0.9$
- The environment is deterministic
- The terminal state s_9 has value $V(s_9) = 0$ (no further rewards after termination)

Part 1: Q-value Calculation

Assume the agent follows a uniform random policy: $\pi(a|s) = 0.25$ for all $a \in \mathcal{A}$ in all non-terminal states.

1. Compute $Q(s_5, \text{Right})$ using the Bellman expectation equation:

$$Q^\pi(s, a) = \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V^\pi(s')]$$

where $V^\pi(s')$ is given by the following current value function estimates (after some iterations of policy evaluation):

$V(s_1) = 2.1$	$V(s_2) = 3.5$	$V(s_3) = 5.0$
$V(s_4) = 1.8$	$V(s_5) = 4.2$	$V(s_6) = 6.8$
$V(s_7) = 0.5$	$V(s_8) = 2.9$	$V(s_9) = 0$

2. Compute $Q(s_2, \text{Down})$ using the same value function.

Part 2: State Value Calculation

Now consider a different policy where:

- In s_5 : $\pi(\text{Right}|s_5) = 0.7$, $\pi(\text{Down}|s_5) = 0.3$, $\pi(\text{Up}|s_5) = 0$, $\pi(\text{Left}|s_5) = 0$
- In s_8 : $\pi(\text{Right}|s_8) = 0.6$, $\pi(\text{Up}|s_8) = 0.4$, $\pi(\text{Down}|s_8) = 0$, $\pi(\text{Left}|s_8) = 0$
- Assume deterministic transitions with rewards as specified earlier

The Q-values for these states under some policy are given as:

State-Action pair	Q-value
$Q(s_5, \text{Right})$	8.2
$Q(s_5, \text{Down})$	6.5
$Q(s_8, \text{Right})$	9.8
$Q(s_8, \text{Up})$	7.3

1. Compute $V^\pi(s_5)$ using the policy-weighted average:

$$V^\pi(s) = \sum_{a \in \mathcal{A}} \pi(a|s) Q^\pi(s, a)$$

2. Compute $V^\pi(s_8)$ using the same method.

Part 3: Optimal Values

Assuming the agent finds the optimal policy π^* :

1. Write the Bellman optimality equation for $Q^*(s, a)$.
2. If $Q^*(s_6, \text{Right}) = 9.0$ and all other actions from s_6 have lower Q-values, what is $V^*(s_6)$?
3. If $V^*(s_2) = 8.4$ and taking "Right" from s_2 leads deterministically to s_3 with $V^*(s_3) = 9.0$, what is $Q^*(s_2, \text{Right})$? (Use the reward of -1 for this transition)

2 Mountain car

The environment is described in https://gymnasium.farama.org/environments/classic_control/mountain_car/. Find the optimal policy by using neural network and following algorithm

1. Deep Q learning
2. Vanilla policy gradient.
3. Actor-Critic.
4. PPO.